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Academic Positions

Aug. 2024 – Now 📖 **PostDoctoral Fellow**
Gran Sasso Science Institute, L'Aquila (AQ), Italy.
Supervisor: Prof. Catia Trubiani
Research Project: DREAM - PRIN 2022

Education

Nov. 2020 – Jul. 2024 📖 **Ph.D. in Computer Science & Mathematics**
University of Camerino, Camerino (MC) Italy Thesis Title: Adaptive Digital Twin Supporting IoT-Enhanced Business Process.
Supervisor: Prof. Barbara Re.

Oct. 2017 – May 2020 📖 **M.Sc. Computer Science LM-18, University of Camerino**
Thesis Title: Modeling and Enactment of Smart Environments combining Business Process and Internet of Things.
Mark: 110/110 cum laude.

Oct. 2014 – Jan. 2018 📖 **B.Sc. Computer Science L-31, University of Camerino**
Thesis Title: Data Mining in the Management of a Non-Profit Organisation.
Mark: 110/110 cum laude.

Visiting Experiences

Mar. 2024 – Jul. 2024 📖 **Visiting Researcher** (5 Months)
Research stay at the **Information Systems Engineering Research Group (LIRIS)** at **KU Leuven**, Brussels, under the supervision of Prof. Monique Snoeck and Prof. Estefania Serral Asensio. During the research stay, I worked on Digital Twins [13] and Learning Analytics in BPM [14].

Research Projects

- 📖 **DREAM - PRIN 2022.** Member of the founded national project **DREAM** modular software Design to Reduce uncertainty in Ethics-based cyber-physical systems. PNRR - PRIN 2022.
Supervisor: Prof. Catia Trubiani
Duration: 24 months
- 📖 **ALMONDO - PRIN 2022.** Member of the founded national project **ALMONDO** Develop a novel modeling framework to study climate lobbying that carefully considers the features of human learning and communication. PNRR - PRIN 2022.
Supervisor: Dr. Fabrizio Fornari
Duration: 12 months

Research Projects (continued)

- **FLUIDWARE - PRIN 2017.** Member of the founded national project **FLUIDWARE** Develop a novel programming model for IoT services and applications. MIUR - PRIN 2017.
Supervisor: Prof. Barbara Re
Duration: 36 months

Research Focus

- SE for AI/GenAI** ■ Software architectures and performance evaluation for AI-based systems, with a particular focus on distributed learning systems. My work examines and evaluates how architectural design choices and runtime conditions influence both system-level properties, such as training time, communication efficiency, and resource usage, and learning-level properties, such as model accuracy. I also work on developing architectural and software engineering solutions, including architectural patterns, benchmarking frameworks, runtime adaptation mechanisms, and agentic solutions, to improve the efficiency, robustness, and overall quality of AI-based systems.
- Business Process Management** ■ Empirical studies on business process modeling, with particular attention to the BPMN standard and its use in practice. My research focuses on large-scale analyses of BPMN models to investigate notation usage, practical complexity, element combinations, syntactic issues, and modeling practices. It also includes the development of tools for automated inspection and quantitative characterization of BPMN models.
- IoT and Digital Twins** ■ Modeling and analysis of IoT-Enhanced Business Processes and Digital Twins, with particular attention to processes in which IoT devices, such as sensors and actuators, are integrated into process execution. My work focuses on conceptual architectures and model-based solutions for process monitoring, what-if analysis, and the alignment between physical devices and their digital representations through Digital Process Twins.

Keywords: AI-based Systems; AI for Software Engineering; Performance Evaluation; Software Architecture; Self-Adaptive Systems; Architectural Patterns; Distributed Learning; Federated Learning; Business Process Management; BPMN; Process Modeling; Process Analysis; Internet of Things; IoT-Enhanced Business Processes; Digital Twins; Digital Process Twins; Conceptual Modeling.

Research Activities

Organizing Committee

- Workshop ■ **Co-Chair.** 3rd International Workshop on Software Architecture for Generative AI (SAGAI) at **ICSA27** [\[Link\]](#)
- **Co-Chair.** 1st International Workshop on Architecting Agentic AI Systems (AgentArch) at **ICSA27** [\[Link\]](#)
- **Co-Chair.** 2nd International Workshop on Software Architecture for Generative AI (SAGAI) at **ICSA26** [\[Link\]](#)

Research Activities (continued)

- Symposium  **Program Co-Chair.** 14th International Symposium DataMod: From Data to Models and Back (DataMod) at **SEFM'26** [\[Link\]](#)
- Special Issue  **Guest Editor.** *Innovations in software architecture using agentic AI: Design, evaluation and runtime solutions*
Special Issue in **Information and Software Technology** [\[Link\]](#)

Program Committee

- Conference  **PC-Member.** International Conference on Software Architecture (ICSA) at **ICSA'27** [\[Link\]](#)
-  **PC-Member.** International Conference on Software Analysis, Evolution and Reengineering (SANER) at **SANER'27** [\[Link\]](#)
- Workshop  **PC-Member.** 2nd IEEE Big Data International Workshop on Software Engineering for Agentic AI (SE4AgenticAI) at **IEEE BigData'26** [\[Link\]](#)
-  **PC-Member.** 4th International Workshop on Reliable and Secure AI for Software Engineering (ReSAISE) at **ISSRE'26** [\[Link\]](#)
-  **PC-Member.** 8th International Workshop on Automated and verifiable Software sYstem DEvelopment (ASYDE) at **ASE'26** [\[Link\]](#)
-  **PC-Member.** 4th International Workshop on Business Processes Meet the Internet of Things (BP-Meet-IoT) at **CBI/EDOC'26** [\[Link\]](#)
-  **PC-Member.** International Conference on Business Informatics & Enterprise Design, Operations and Computing: Tools & Demos Track (CBI & EDOC) at **CBI & EDOC'26** [\[Link\]](#)
-  **PC-Member.** 4th International Workshop on Modelling and Implementation of Digital Twins for Complex Systems (MIDas4CS) at **CAiSE'26** [\[Link\]](#)
-  **PC-Member.** 1st International Workshop on Engineering Autonomous Systems Intelligence (EASI) at **CAiSE'26** [\[Link\]](#)
-  **PC-Member.** 1st International Workshop on AI Code QUality, Integrity & RELiability (ACQUIRE) at **EDCC'26** [\[Link\]](#)
-  **PC-Member.** 1st International Workshop on Software Architecture for Data-Intensive Systems (SADIS) at **ECSA'25** [\[Link\]](#)

Review Activities

- Q₁  **Journal of Systems and Software** [\[Link\]](#)
-  **Future Generation Computer Systems** [\[Link\]](#)
-  **Information and Software Technology** [\[Link\]](#)
-  **Information Fusion** [\[Link\]](#)
-  **Ad Hoc Networks** [\[Link\]](#)
-  **Internet of Things** [\[Link\]](#)
- Q₂  **Displays** [\[Link\]](#)
-  **Cogent Business & Management** [\[Link\]](#)
- Q₃  **Science of Computer Programming** [\[Link\]](#)
-  **SoftwareX** [\[Link\]](#)

Awards

- 2023  **BPM 2023 Best Paper Award (Demonstration and Resources Forum)**
Award received for the paper *BPMN Inspector: A Tool for Extracting Features from BPMN Models* [12], presented at the 21st International Conference on Business Process Management (BPM 2023). <https://bpm-conference.org/awards/>
The work was also highlighted in the November 2023 issue of the BPM Newsletter for its potential impact on BPM practitioners and educators. <https://bpm-conference.org/assets/docs/newsletter/BPM-newsletter-2023-11.pdf>

Teaching and Tutoring

- 2024  **STEM, Digital and Innovation Skills – Teaching** (for Primary School)
Tutor in the PNRR project “Crescere per scoprire e aprirsi al mondo” (Mission 4 – Education and Research, Investment 3.1 “New Skills and New Languages”, D.M. n. 65/2023), delivering educational and orientation activities aimed at strengthening STEM, digital, and innovation skills, with a focus on inclusion and equal opportunities.
Project code: M4C1I3.1-2023-1143-P-30633. C.U.P. G84D23005080006.
- 2021 - 2023  **Operating Systems – Tutoring** (for M.Sc. Students)
I served as a tutor for the Operating Systems course at the University of Camerino during the academic years 2021/2022 and 2022/2023.
- 2020 - 2024  **Thesis Supervision**
During my Ph.D., I supervised 5+ bachelor’s theses in the Computer Science program at the University of Camerino.
-  **Third Mission Activities**
I was involved in public engagement activities in schools, delivering lessons and workshops to students from kindergarten to upper secondary school. These activities focused on programming and computer science, with content adapted to different age groups. This activity reflects the university’s societal impact mission to support the goals of the Italian PNRR to promote STEM and digital skills across all levels of education.

Contributed Talks

- 2026  **[Participant]** 23rd International Conference on Principles of Knowledge Representation and Reasoning (**KR ’26**), Lisbon, Portugal, July 20–23, 2026.
-  **[Speaker]** **Towards Agentic Runtime Reconfiguration of Architectural Patterns in Federated Learning**
Seminar for SPEC, Online, June 1, 2026.
-  **[Speaker]** 21st International Conference on Software Engineering for Adaptive and Self-Managing Systems (**SEAMS’26**) co-located with the International Conference on Software Engineering **ICSE’26**, Rio de Janeiro, Brazil, April 12–18, 2026.
- 2025  **[Speaker]** 4th Conference on System and Service Quality, (**QualITA’25**), Catania, Italy, June 25 and 27, 2025.
-  **[Speaker]** **Experimenting Architectural Patterns in Federated Learning Systems**
Seminar at Gran Sasso Science Institute, L’Aquila, 16 April 2025.
-  **[Speaker]** 22nd IEEE International Conference on Software Architecture (**ICSA ’25**), Odense, Denmark, 31 March – April 4, 2025.

Contributed Talks (continued)

- 2024  **[Speaker]** **Trends on the use and adoption of the BPMN during the last ten years**
Seminar at the University of Trieste, Trieste, 18 December 2024.
-  **[Speaker]** **[Session Chair]** 28th International Conference on Enterprise Design, Operations, and Computing (EDOC '24), Vienna, Austria, September 10–13, 2024.
-  **[Speaker]** **Digital Process Twin: Open Challenges**
Seminar at Gran Sasso Science Institute, L'Aquila, 17 July 2024.
- 2023  **[Speaker]** 26th International Conference on Model-Driven Engineering Languages and Systems (MODELS '23), Västerås, Sweden, October 1–6, 2023.
-  **[Speaker]** 21st International Conference on Business Process Management (BPM '23), Utrecht, The Netherlands, September 11–15, 2023.
-  **[Speaker]** **Trends on the adoption of BPMN during the last decade**
Seminar at the University of Camerino, Camerino, 2023.
- 2022  **[Speaker]** **The use of the BPMN in IoT Environments**
Seminar at the University of Camerino, Camerino, 2022.
- 2021  **[Speaker]** 20th international Conference on Business Informatics Research (BIR '21), Vienna, Austria, September 22–24, 2021.
- 2020  **[Participant]** 18th International Conference on Business Process Management (BPM '20), Seville, Spain, September 13–18, 2020.

Research Publications

Journal Articles

-  **J1** **Compagnucci, I.**, Lu, Q., & Trubiani, C. (2026). Agentic Runtime Reconfiguration of Architectural Patterns in Federated Learning. *Journal of Systems and Software (JSS)*, 240, 112966.
 doi:10.1016/j.jss.2026.112966
-  **J2** **Compagnucci, I.**, Pincirolì, R., & Trubiani, C. (2026). Experimenting Architectural Patterns in Federated Learning Systems. *Journal of Systems and Software (JSS)*, 232, 112655.
 doi:10.1016/J.JSS.2025.112655
-  **J3** Fornari, F., **Compagnucci I.**, Donato, M. C. D., Bertrand, Y., Beyel, H. H., Carrión, E., ... Valderas, P. (2025). Digital Twins of Business Processes: A Research Manifesto. *Internet of Things*, 30, 101477.
 doi:10.1016/J.IOT.2024.101477
-  **J4** **Compagnucci, I.**, Corradini, F., Fornari, F., & Re, B. (2024). A Study on the Usage of the BPMN Notation for Designing Process Collaboration, Choreography, and Conversation Models. *Business & Information Systems Engineering, (BISE)*, 66, 43–66.  doi:10.1007/S12599-023-00818-7
-  **J5** **Compagnucci, I.**, Corradini, F., Fornari, F., Polini, A., Re, B., & Tiezzi, F. (2022). A Systematic Literature Review on IoT-Aware Business Process Modeling Views, Requirements and Notations. *Software and Systems Modeling, (SoSyM)*, 14(1), 1–36.  doi:10.1007/S10270-022-01049-2

Conference Proceedings

-  **C1** Silvetti, S., **Compagnucci, I.**, Cairolì, F., Trubiani, C., & Nenzi, L. (2026). Time Robustness for Point-Based Semantics of Metric Interval Temporal Logic. In *Proceedings of the 23rd International Conference on Principles of Knowledge Representation and Reasoning (KR)* (pp. 589–599).
 doi:10.24963/kr.2026/56

- C2** Baresi, L., **Compagnucci, I.**, Lestingi, L., & Trubiani, C. (2026). Adaptive Toggling of Architectural Patterns for Federated Learning. In *Proceedings of the 21st international conference on software engineering for adaptive and self-managing systems (SEAMS)* (pp. 105–116).  doi:10.1145/3788550.3794873
- C3** Kurkchi, Y., **Compagnucci, I.**, & Trubiani, C. (2026). Understanding the Concepts behind Ethics in Designing Cyber-Physical Systems. In *19th International Conference on the Quality of Information and Communications Technology*.
- C4** **Compagnucci, I.**, Pincirolì, R., & Trubiani, C. (2025). Performance Analysis of Architectural Patterns for Federated Learning Systems. In *International Conference on Software Architecture, (ICSA)* (pp. 289–300).  doi:10.1109/ICSA65012.2025.00036
- C5** **Compagnucci, I.**, & Trubiani, C. (2025). Towards AI Agents for Selecting Architectural Patterns in Federated Learning Systems. In *4th Conference on System and Service Quality (QualITA 2025)*. (Vol. 4080).
- C6** **Compagnucci I.**, Re, B., Asensio, E. S., & Snoeck, M. (2024). A Digital Process Twin Conceptual Architecture for What-If Process Analysis. In *Enterprise Design, Operations, and Computing. (EDOC) 2024 Workshops* (Vol. 537, pp. 373–388).  doi:10.1007/978-3-031-79059-1_23
- C7** **Compagnucci, I.**, Corradini, F., Fornari, F., & Re, B. (2023). BPMN Inspector: A Tool for Extracting Features from BPMN Models. In *Proceedings of the Best Dissertation Award, Doctoral Consortium, and Demonstration & Resources Forum at (BPM)* (Vol. 3469, pp. 122–126).
- C8** **Compagnucci, I.**, Snoeck, M., & Asensio, E. S. (2023). Supporting Digital Twins Systems Integrating the MERODE Approach. In *International Conference on Model Driven Engineering Languages and Systems: Companion Proceedings, MODELS-C* (pp. 449–458).  doi:10.1109/MODELS-C59198.2023.00079
- C9** Vemuri, P., Poelmans, S., **Compagnucci, I.**, & Snoeck, M. (2023). Using Formative Assessment and Feedback to Train Novice Modelers in Business Process Modeling. In *International Conference on Model Driven Engineering Languages and Systems: Companion Proceedings, MODELS-C* (pp. 449–458).  doi:10.1109/MODELS-C59198.2023.00079
- C10** **Compagnucci, I.**, Corradini, F., Fornari, F., & Re, B. (2021). Trends on the Usage of BPMN 2.0 from Publicly Available Repositories. In *International Conference on Business Informatics Research, (BIR)* (Vol. 430, pp. 84–99).  doi:10.1007/978-3-030-87205-2_6
- C11** **Compagnucci, I.**, Corradini, F., Fornari, F., Polini, A., Re, B., & Tiezzi, F. (2020). Modelling Notations for IoT-Aware Business Processes: A Systematic Literature Review. In *Business Process Management Workshops (BPM)* (Vol. 397, pp. 108–121).  doi:10.1007/978-3-030-66498-5_9

Software

BPMN INSPECTOR A Tool for Extracting Features from BPMN Models

Version. 1.0.0

■ BPMN INSPECTOR for the analysis of collections of BPMN models. It supports research by automatically analyzing large collections of BPMN models and providing quantitative data on their characteristics. More specifically, it classifies models by type, detects duplicates and invalid models, and flags non-English models, thereby removing the need for manual inspection of these aspects. It also provides a detailed characterization of the analyzed collection by quantifying the use of BPMN notation elements and their combinations, identifying syntactic violations of the standard, and assessing if the models comply with established good modeling practices.

Link: <https://github.com/PROSLab/BPMN-Inspector>

Related References: [4, 12, 15]

 **BPM 2023 Best Paper Award (Demonstration and Resources Forum)** [\[Link\]](#)

Software (continued)

AP4FED: A Federated Learning Benchmarking Platform

Version 1.5.0

■ AP4FED is a Federated Learning platform built on top of Flower that supports the execution, monitoring, and adaptive configuration of Federated Learning processes. It extends Flower with a set of architectural patterns that can be dynamically enabled or disabled during training, allowing the system to adapt the execution strategy to the current conditions of the learning process. In particular, AP4FED supports adaptive decisions on patterns such as client selection, heterogeneous data handling, and message compression, aiming to improve performance while introducing only limited overhead. It also provides monitoring capabilities that collect execution data and enable the evaluation of different adaptation strategies in experimental settings.

Link: <https://github.com/IvanComp/AP4Fed>

Related References: [7, 2, 9]